GoPaperless Payment Orchestration Architecture

Multi-Provider Async Payment Engine

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1 — FRONTEND LAYER · WEB / MOBILE RUNTIME

usePayment

- Start / retry / resume
- Poll payment status
- Provider locking
- Offline detection

Runtime core

usePaymentRouting

- shouldStartPolling()
- isCallbackFlow()
- isMobileDirectFlow()
- requiresMnoSelection()

FE prediction only

usePaymentRecovery

- checkRecovery()
- resumeRecovery()
- discardRecovery()
- activeReference restore

Hydration recovery

PaymentStep.vue

- Phone input + E164
- MNO detection UX
- Provider selection chips
- Resumable polling UX

UX orchestrator

MNO detection states: idle → detecting → detected / suggested / requires-selection → selected. Provider locked once reference acquired.

HTTP / REST boundary

2 — API LAYER · PAYMENTCONTROLLER

Endpoints

```
POST /payment/start
POST /payment/charge-mobile
GET /payment/status
GET /payment/verify
GET /payment/mobile-options
POST /payment/callback/daraja
POST /payment/callback/dpo
```

Controller contract

- Thin orchestration boundary
- Delegates all logic to PaymentService
- No routing ownership
- No retry ownership
- No persistence ownership

Passthrough only

Service boundary

3 — APPLICATION LAYER · PAYMENTS ENGINE

PaymentService

Core orchestration authority

- startPayment() — idempotency + routing
- getPaymentStatus() — DB read-only
- resumePayment() — hydration safe
- verifyPayment() — one-shot UX boost
- syncPaymentStatus() — BG job only
- chargeMobile() — DPO deferred charge
- queryPayment() — Daraja fallback
- handleDarajaCallback()
- handleDpoCallback()

Execution authority

Sub-services

MobileMoneyService

- initiate() — route to Daraja / DPO
- charge() — DPO deferred only
- getMobileOptions()

CardService

- initiateCard() → DPO redirect

VerificationService

- verifyStatus() — DPO polling
- query() — Daraja fallback

FxEngineService

- convert() — KES → local currency
- identityResult() — card bypass

ProviderExecutionState

Lifecycle state machine

INITIATED · REQUIRES_SELECTION

EXECUTING · RECONCILING · SUCCESS

FAILED · CANCELLED · INVALID_REQUEST

Metadata DTOs

- PaymentMetadataDTO
- ProviderPayloadDTO
- SelectedMnoDTO
- SuggestedMnoDTO
- SelectionDTO

Execution state ≠ business state

Domain intelligence boundary

4 — DOMAIN INTELLIGENCE LAYER · MOBILE MONEY DECISION PIPELINE

PhoneResolutionService

- libphonenumber (single entry)
- E164 normalisation
- Region extraction (ISO)
- Carrier hint (best effort)
- Returns PaymentPhoneResolutionDTO

Phone → region

MnoDetectionRegistry

- KE, TZ, UG, RW, MW, ZM, ZW
- Prefix → MNO mapping
- Ambiguous prefix flagging
- Multi-match portability detection

HIGH · AMBIGUOUS

UNKNOWN

MnoRoutingRegistry

- route() — capability + region + carrier + confidence → MnoRoutingResultDTO
- supportsRegion()
- supportedRegions()
- mnoKey resolution

Routes to provider

PaymentCountryResolver + FxEngine

- region → country, currency, min/max
- KES → local currency conversion
- FX rate lock at payment creation
- Card: identity FX (bypass)
- Daraja: always KES

Country + FX context

PHONE RESOLUTION PIPELINE

Phone input (+E164) → libphonenumber parse → Region (ISO) → CountryResolver → currency, limits → MnoDetectionRegistry → confidence → MnoRoutingRegistry → provider + mnoKey → FxEngine → display amount

AMBIGUOUS or UNKNOWN confidence → requiresUserSelection=true → DPO GetMobilePaymentOptions fetched → chip selector shown to user

Provider adapter boundary — no orchestration crosses here

5 — PROVIDER LAYER · ADAPTERS

DpoProvider

- initiateCard() — CreateTransaction token
- initiateMobileMoney() — CreateTransaction token
- charge() — ChargeTokenMobile
- getMobileOptions() — GetMobilePaymentOptions
- test()

Execution only · No routing · No retry · No persistence

DarajaProvider

- initiateMobileMoney() — STK Push
- query() — MpesaExpressQuery fallback
- test()

Callback-only provider. No polling flow. Query used as fallback when callback delayed (>20s).

Execution only · No routing · No retry · No persistence

External network boundary

6 — EXTERNAL PROVIDERS · APIS

DPO Group

- Card payments — redirect flow
- Mobile money — mobile direct flow
- ChargeTokenMobile (deferred charge)
- GetMobilePaymentOptions (provider list)
- Callback → /payment/callback/dpo

DPO Pay API

Safaricom Daraja

- STK Push → callback flow
- CheckoutRequestID as reference
- Callback → /payment/callback/daraja
- MpesaExpressQuery fallback

Daraja API 2.0

Persistence boundary

7 — PERSISTENCE LAYER · GOPAPERLESS_PAYMENTS

Payment entity — core fields

- paymentAttemptId (idempotency)
- providerReference (lookup key)
- paymentStatus (business state)
- provider / paymentProvider
- phoneE164Digits / region / country
- amount / currency (minor units)
- displayAmount / displayCurrency
- fxRate / fxRateSource / fxRateLockedAt

Orchestration metadata — JSON blob

- providerExecutionState
- executedProvider
- flow (redirect | callback | mobile_direct)
- method (mobile | card)
- selected / suggested / selection
- confidence
- alreadyCharged
- instructions
- providerPayload (init + charge + callback + query snapshots)
- providerError
- context (carrier, region, confidence breakdown)

Metadata is first-class orchestration state

Verification + retry fields

- verificationStatus
- verificationRetryCount
- verificationLastCheckedAt
- nextVerificationAt
- verificationLockedAt (BG job mutex)
- createdAt (expiry reference)
- paidAt
- lastErrorCode / lastErrorMessage

15 min expiry window

Data flow examples

8 — DATA FLOWS

Flow 1 · Card / Redirect

FE startPayment(card) → CardService → DPO CreateTransaction → redirectUrl returned

Flow 2 · Mobile Direct (DPO)

Phone → PhoneResolution → Region → MnoDetection → confidence → MnoRouting →

Flow 3 · Callback (Daraja)

Phone → KE → Daraja STK Push → FE starts polling DB Status immediately

→ FE redirect

User completes card form → DPO callback

handleDpoCallback() → finalisePayment()

verifyPayment() one-shot → PAID

DPO CreateTransaction → reference

FE: HIGH → auto / AMBIGUOUS → chips / UNKNOWN → selector

chargeMobile() → ChargeTokenMobile

Poll /status → DPO verify → PAID

Daraja callback arrives → handleDarajaCallback()

finalisePayment() → PAID + entitlement

If callback delayed >20s → queryPayment() fallback

Callback is source of truth. Query is fallback only. Never fail hard on query error.

9 — RECOVERY & RESUMABILITY

Resumable orchestration

- resumePayment() → hydrates persisted session
- activeReference restored from DB
- Provider locked — prevents orchestration drift
- MNO / selection state restored
- FX context restored for display
- Polling resumes if alreadyCharged=true

Payments are resumable — but not indefinitely resumable

Bounded recovery semantics

- 15 min expiry window from createdAt
- Expired → status set FAILED, recovery stopped
- Expiry is UX boundary, NOT provider settlement truth
- Expired payment may still reconcile via async callback
- hasPaymentExpired() checked at: status, verify, sync, resume
- BG job stops retrying after expiry

10 — FINALITY & RECONCILIATION

Reconciliation layers

- Provider callback (primary)
- FE one-shot verify (UX boost)
- BG job polling + retry scheduling
- Daraja query fallback (>20s)

Provider acknowledgement ≠ durable success

Idempotency guarantees

- paymentAttemptId prevents duplicates
- Duplicate callbacks silently ignored
- finalisePayment() checks PAID before tx
- Atomic DB tx: payment update + entitlement creation

DB transaction wraps finality

Final immutable states

PAID FAILED EXPIRED

INITIATION_FAILED also terminal. Background job cannot resurrect any terminal state.

11 — FUTURE ARCHITECTURE (ROADMAP)

Formal FSM + transition guards | Reconciliation engine | Weighted provider routing | Provider failover | Provider health scoring | Generic transaction platform | Entitlement ledger | Transition engines | Multi-currency settlement

12 — KEY ARCHITECTURAL PRINCIPLES

- **Backend is execution authority**
- **FE predicts** — backend decides
- **Provider locking** prevents orchestration drift
- **Orchestration must survive** refreshes
- **Recovery is bounded** — 15 min window
- **Adapters are execution-only**
- **Execution state ≠ business state**
- **Metadata is first-class** orchestration state
- **Amounts stored in minor units** — converted at boundary
- **FX locked at payment creation** — never re-fetched
- **One entitlement per payment** — atomic transaction
- **Idempotency enforced** at all callback ingress